

CONSTRUCTOR :

A constructor is a special method used to initialize objects.

A constructor is called when an instance of a class is created. It has the same name as the class and does not have a return type, not even void.

Syntax to create the constructor :

A programmer can define a constructor by using the following syntax :

```
access_modifier ClassName([Formal_Arguments])
{
    // logic of initialization
}
```

PURPOSE OF THE CONSTRUCTOR :

During the execution of the constructor,

- Non-static members of the class will be loaded into the object.
- Programmer written instruction of the constructor gets executed.

Key Characteristics of Constructors

*Name: The constructor must have the same name as the class.

*No Return Type: Constructors do not have a return type.

*Overloading: Constructors can be overloaded by defining multiple constructors with different parameter lists.

Types of Constructors

1.Default Constructor: If no constructor is defined in a class, the compiler automatically creates a default constructor with no arguments. This constructor initializes object fields with default values.

// Refer class Notes and Programs for Examples.

2.User-defined constructor

A user-defined constructor in Java is a constructor that is explicitly defined by the programmer. User-defined constructors allow you to initialize objects with specific values or execute custom initialization logic when an object is created.

No-argument constructor

A no-argument constructor in Java is a constructor that does not take any parameters/ arguments. It is used to create an object with default values.

Parameterized Constructor:

A constructor that accepts arguments to initialize an object with specific values.

// Refer class Notes and Programs for Examples.

Constructor Overloading

we can have multiple constructors in a class with different parameter lists. This is known as constructor overloading.

```
public class Student {
```

```
    int id;
```

```
    String name;
```

```
    int age;
```

```
    // constructor
```

```
// Non - Parameterized Constructor:
```

```
public Student()
```

```
{
```

```
}
```

```
// Parameterized Constructor:
```

```
public Student(int id ,String name , int age )
```

```
{
```

```
    this.id = id;
```

```
    this.name = name;
```

```
    this.age = age;
```

```
}
```

```
}
```